

Problem 11.18

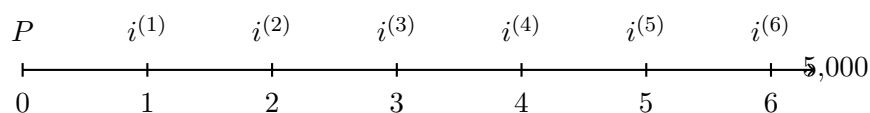
A fund pays a nominal rate of 12%. The nominal rate is compounded once in year 1, twice in year 2, three times in year 3, etc. Calculate the amount that must be invested today in order to accumulate 5,000 after 6 years.

Solution

Let P denote the amount invested today. In year k , the nominal annual rate $i^{(k)} = 0.12$ is compounded k times, so the accumulation factor for that year is

$$\left(1 + \frac{0.12}{k}\right)^k.$$

The six-year timeline is:



Therefore,

$$\begin{aligned} 5,000 &= P \prod_{k=1}^6 \left(1 + \frac{0.12}{k}\right)^k, \\ P &= \frac{5,000}{(1 + 0.12) \left(1 + \frac{0.12}{2}\right)^2 \left(1 + \frac{0.12}{3}\right)^3 \left(1 + \frac{0.12}{4}\right)^4 \left(1 + \frac{0.12}{5}\right)^5 \left(1 + \frac{0.12}{6}\right)^6} \\ &= 2475.0872937 \dots \end{aligned}$$

Thus, the amount that must be invested today is

$$\boxed{P \approx 2,475.09}.$$